EULER'S PENTAGONAL NUMBER THEOREM-TYPE RESULTS VIA MINIMAL EXCLUDANTS OF PARTITIONS AND OVERPARTITIONS

ARITRAM DHAR, RAJAT GUPTA, AVI MUKHOPADHYAY, AND RISHABH SARMA

ABSTRACT. We investigate weighted sums involving the minimal excludant statistic on partitions, overpartitions, and certain restricted classes. This study leads to interesting results analogous to the Euler's pentagonal number theorem.

1. INTRODUCTION

One of the most fundamental theorems in the theory of partitions is the Euler's pentagonal number theorem, which states that,

$$(q;q)_\infty = 1 + \sum_{k=1}^{\infty} (-1)^k \left(q^{\frac{k(3k-1)}{2}} + q^{\frac{k(3k+1)}{2}} \right),$$

where $(A;q)_N = (1-A)(1-Aq)\cdots(1-Aq^{N-1})$ for $N \in \mathbb{N} \cup \{\infty\}$ and $(A;q)_0 = 1$ is the usual q-Pochhammer symbol. The celebrated involution of Franklin gives a combinatorial interpretation of the relation above in the form of the following weighted identity for $n \geq 1$,

$$\sum_{\pi \in D(n)} (-1)^{\#(\pi)} = \begin{cases} (-1)^k, & \text{if } n = \frac{k(3k\pm1)}{2}, \\ 0, & \text{otherwise,} \end{cases}$$

where $\#(\pi)$ denotes the number of parts of π. Subsequently, Andrews [2] in 1972 gave beautiful Franklin type combinatorial proofs of the following two identities due to Gauss [12, 8; p. 447, Eqs. (14) and (10)]

$$\frac{(q^2;q^2)_\infty}{(-q;q^2)_\infty} = 1 + \sum_{n=1}^{\infty} (-1)^n q^{n(2n-1)} \left(1 + q^{2n}\right), \tag{1.1}$$

and

$$\frac{(q;q)_\infty}{(-q;q)_\infty} = 1 + 2\sum_{n=1}^{\infty} (-1)^n q^{n^2}. \tag{1.2}$$

Following the same definition as Andrews, define $G(n)$ be the set of all partitions of n in which no even parts are repeated. He proved the following weighted type identity from (1.1).

Theorem 1.1. [2, Theorem 1] *For $n \geq 1$, we have*

$$\sum_{\pi \in G(n)} (-1)^{\#(\pi)} = \begin{cases} (-1)^k & \text{if } n = k(2k \pm 1), \\ 0 & \text{otherwise.} \end{cases}$$

Date: April 9, 2026.

2020 *Mathematics Subject Classification.* 05A15, 05A17, 11P81.

Key words and phrases. Distinct parts, minimal excludant, overpartitions, partitions, q-series, weighted sums.

Similarly, if $g(n)$ is the set of all partitions of n of the form

$$n = \sum_{i=1}^{s} a_i + \sum_{j=1}^{t} b_j,$$

with $a_i > a_{i+1}$, $b_j \geq b_{j+1}$. Then Andrews proved the following weighted identity equivalent to (1.2).

Theorem 1.2. [2, Theorem 2] *For* $n \geq 1$,

$$\sum_{\pi \in g(n)} (-1)^{\#(\pi)} = \begin{cases} 2(-1)^k & \textit{if } n = k^2, \\ 0 & \textit{otherwise}. \end{cases}$$

Note that the numbers appearing in Andrews' results are hexagonal and square numbers. Results of this type have subsequently appeared in literature and been studied by several authors in the past decades. For instance, Alladi [1, p. 228, Theorem 2] studied the set of partitions of a non-negative integer into distinct parts with the smallest part being odd, denoted by $P_{d,o}(n)$ and proved that for $n \geq 1$,

$$\sum_{\lambda \in P_{d,o}(n)} (-1)^{\#(\pi)} a^{\#_{odd}(\pi)} = \begin{cases} (-a)^k & \text{if } n = k^2, \\ 0 & \text{otherwise}, \end{cases}$$

where $\#_{odd}(\pi)$ denote the number of odd parts in a partition π.

In this paper, we study weighted identities over sets of overpartitions where the weight is the mex statistic studied by Aricheta and Donato [6] (see the discussion above (1.5)). Using this framework, we establish infinite families of weighted identities whose generating functions exhibit a structure analogous to Euler's pentagonal number theorem. In particular, we employ a two-variable generalization of the mex statistic for overpartitions to achieve our goal.

We begin by recalling the origin and subsequent developments of the mex statistic. In 2019, Andrews and Newman [5] undertook a combinatorial study of the minimal excludant or "mex" of an integer partition, originally introduced by Grabner and Knopfmacher [13] under the name "smallest gap". The work of Andrews and Newman initiated the study of the mex, its generalizations [9], extensions to overpartitions [6, 10] and their connections to other partition theoretic objects and statistics [9, 15]. For an integer partition π, the statistic mex(π) is defined as the smallest positive integer that is not a part of π. For instance, if we take $\pi = 8 + 5 + 2 + 1$ is a partition of 16, then $\text{mex}(\pi) = 3$.

In [5], Andrews and Newman initiated the study of the weighted sum of the mex statistic taken over all partitions of n. More precisely, they defined

$$\sigma\text{mex}(n) := \sum_{\pi \in \mathcal{P}(n)} \text{mex}(\pi), \tag{1.3}$$

where $\mathcal{P}(n)$ denotes the set of all partitions of n. They proved [5, Theorem 1.1] that

$$\sigma\text{mex}(n) = D_2(n),$$

where $D_2(n)$ denotes the number of partitions of n into distinct parts with two colors.

A related study involving the weighted sum of the mex statistic restricted to partitions into distinct parts was carried out by Kaur et al. [16]. In this setting, the corresponding sum, denoted by $\sigma_d\text{mex}(n)$, is defined by summing the mex values over all partitions of n into distinct parts. They showed that for

$$\sigma_d\text{mex}(n) := \sum_{\pi \in \mathcal{D}(n)} \text{mex}(\pi), \tag{1.4}$$

where $\mathcal{D}(n)$ is the set of distinct partitions of n, we have

$$\sum \sigma_d\text{mex}(n) q^n = R(q)$$

where

$$R(q) := \sum_{n=0}^{\infty} \frac{q^{\frac{n(n+1)}{2}}}{(-q;q)_n}.$$

The function $R(q)$ was introduced by Ramanujan [17]. For further study on $R(q)$, see the works of Andrews [3] and Andrews, Dyson and Hickerson [4].

In 2003, Corteel and Lovejoy [8] introduced overpartitions, which are partitions in which the first (or final) occurrence of a number may be overlined. For example, the eight overpartitions of 3 are 3, $\overline{3}$, $2+1$, $\overline{2}+1$, $2+\overline{1}$, $\overline{2}+\overline{1}$, $1+1+1$, and $\overline{1}+1+1$. Let $\overline{P}(n)$ denote the set of all overpartitions of n.

Recently, Aricheta and Donato [6] extended the concept of minimal excludant to overpartitions. They defined the minimal excludant of an overpartition π, denoted by $\overline{\text{mex}}(\pi)$ to be the smallest positive integer that is not a part of the non-overlined parts of π. Following this, for a positive integer n, the authors define $\sigma\,\overline{\text{mex}}(n)$ to be the sum of the minimal excludants over all overpartitions π of n, namely,

$$\sigma\overline{\text{mex}}(n) := \sum_{\pi \in \overline{P}(n)} \overline{\text{mex}}(\pi) \tag{1.5}$$

and showed that

$$\sigma\overline{\text{mex}}(n) = D_3(n),$$

where $D_3(n)$ is the number of partitions of n into distinct parts using three colors [6, Theorem 2], among other results.

In this paper, we consider two parameter refinements of the three arithmetic functions in Equations (1.3), (1.4) and (1.5). The two variables in each of these refinements keep track of the number of parts and the minimal excludant value of the partitions or overpartitions over which the sum is considered. We define these below.

Throughout, let $c, z \in \mathbb{C}$. Let the quantities $\#_{no}(\pi)$ and $\#_o(\pi)$ denote the non-overlined parts and overlined parts of the overpartition π.

Define

$$\sigma\text{mex}(c,z,n) := \sum_{\pi \in P(n)} c^{\#(\pi)} z^{\text{mex}(\pi)}, \tag{1.6}$$

$$\sigma_d\text{mex}(c,z,n) := \sum_{\pi\in D(n)} c^{\#(\pi)} z^{\text{mex}(\pi)}, \tag{1.7}$$

$$\sigma\overline{\text{mex}}(c,z,n) := \sum_{\pi\in \overline{P}(n)} c^{\#(\pi)} z^{\overline{\text{mex}}(\pi)}, \tag{1.8}$$

$$\sigma_{no}\overline{\text{mex}}(c,z,n) := \sum_{\pi\in \overline{P}(n)} c^{\#_{no}(\pi)} z^{\overline{\text{mex}}(\pi)}, \tag{1.9}$$

and

$$\sigma_{o}\overline{\text{mex}}(c,z,n) := \sum_{\pi\in \overline{P}(n)} c^{\#_{o}(\pi)} z^{\overline{\text{mex}}(\pi)}. \tag{1.10}$$

Note that differentiating (1.8), (1.9), and (1.10) with respect to z, then letting $z \to 1$ and $c \to 1$, we obtain

$$\sigma\overline{\text{mex}}(n) = \sigma_{no}\overline{\text{mex}}(n) = \sigma_{o}\overline{\text{mex}}(n) = \sum_{\pi\in \overline{P}(n)} \overline{\text{mex}}(\pi).$$

We carry out a treatment of the above two variable weighted sums ((1.6)–(1.10)) using combinatorial arguments, hypergeometric transformations and classical analysis to arrive at interesting Pentagonal Number Theorem type results discussed above involving weighted combinations of the minimal excludant values over partitions, overpartitions and their restricted classes. The first of these is quite striking as it equates a combination of mexes over unrestricted and distinct partition classes. We present our results below and prove them in the subsequent section. All of these interestingly involve triangular numbers. The last section is dedicated to further discussions and prospects for future work in this topic.

Our first main result concerns weighted sum taken over partitions of n into distinct parts and weights involving the number of parts of the partition.

Theorem 1.3. *For $|q| < 1$, we have*

$$\sum_{n=1}^{\infty} \left(\sum_{\pi\in D(n)} (-1)^{\#(\pi)} \text{mex}(\pi) \right) q^n = (q;q)_\infty^2.$$

Combining this with [7, Lemma 4.1], namely,

$$\sum_{n=1}^{\infty} \left(\sum_{\pi\in P(n)} (-1)^{\text{mex}(\pi)-1} \text{mex}(\pi) \right) q^n = (q;q)_\infty^2, \tag{1.11}$$

we obtain the following interesting result.

Corollary 1.4. *For every positive integer n, we have*

$$\sum_{\pi\in D(n)} (-1)^{\#(\pi)} \text{mex}(\pi) = \sum_{\pi\in P(n)} (-1)^{\text{mex}(\pi)-1} \text{mex}(\pi).$$

Our second main result concerns weighted sum taken over overpartitions of n and weights involving the number of parts of the overpartition.

Theorem 1.5. *For every positive integer n and non-negative integer ℓ, we have*

$$\sum_{\pi\in\overline{P}(n)}(-1)^{\#(\pi)}\overline{\text{mex}}^{\ell}(\pi)=\sum_{k=0}^{n}a(k)b(n-k),$$

where $a(0)=1$, $a(n)=\sum_{\pi\in\overline{P}(n)}(-1)^{\#(\pi)}$, *and*

$$b(k)=\begin{cases}(-1)^j((1+j)^\ell-j^\ell) & \text{if } k=\frac{j(j+1)}{2}\geq 1,\\ 0 & \text{otherwise}.\end{cases}$$

Setting $\ell=0$, we recover the fundamental result of Corteel and Lovejoy [8, Theorem 4.5]:

$$a(n):=\sum_{\pi\in\overline{P}(n)}(-1)^{\#(\pi)}=\begin{cases}2(-1)^k & \text{if } n=k^2,\\ 0 & \text{otherwise}.\end{cases}$$

Interestingly, this identity had already appeared earlier in the work of Andrews [2, Theorem 2], obtained through a different interpretation of overpartitions (see Theorem 1.2 above). Further taking $\ell=1$ in the above theorem, we have the following corollary.

Corollary 1.6. *For every positive integer n, we have*

$$\sum_{\pi\in\overline{P}(n)}(-1)^{\#(\pi)}\overline{\text{mex}}(\pi)=\sum_{k=0}^{n}a(k)b(n-k),$$

where

$$b(k)=\begin{cases}(-1)^j & \text{if } k=\frac{j(j+1)}{2}\geq 1,\\ 0 & \text{otherwise}.\end{cases}$$

Here we illustrate the case $n=4$ to show the combinatorial beauty of this corollary, or in general, Theorem 1.5. There are 14 overpartitions of 4. Evaluating $(-1)^{\#(\pi)}\overline{\text{mex}}(\pi)$ for each of them and summing the contributions gives

$$\sum_{\pi\in\overline{P}(4)}(-1)^{\#(\pi)}\overline{\text{mex}}(\pi)=-1-1+2+2+1+1+1+1-3-2-3-2+2+2=-2.$$

On the other hand, recall that $a(0)=1$ (empty overpartition), $a(1)=-2$, $a(4)=2$, while $a(2)=a(3)=0$, and $b(1)=-1$, $b(3)=1$, with $b(0)=b(2)=b(4)=0$. Consequently,

$$\sum_{k=0}^{4}a(k)b(4-k)=-2,$$

which agrees with the left-hand side.

Note that the statistic $\overline{\text{mex}}(\pi)$ can be relatively large compared to the parts of the overpartition; for instance, $\overline{\text{mex}}(2,1,1)=3$. Hence the individual terms $(-1)^{\#(\pi)}\overline{\text{mex}}(\pi)$ may have sizeable magnitude. Nevertheless, when summed over all 14 overpartitions of 4, the positive and negative contributions almost completely cancel, leaving the small value -2. This cancellation becomes transparent from the right-hand side. The convolution

$$\sum_{k=0}^{n} a(k)b(n-k)$$

is highly sparse because $a(k)$ is supported only at squares and $b(m)$ only at triangular numbers. Thus contributions occur only when

$$n = k^2 + \frac{j(j+1)}{2}.$$

For $n = 4$, the only such representation is $4 = 1 + 3$, and consequently the sum reduces to a single nonzero term; it would be interesting to obtain a direct combinatorial explanation for this cancellation phenomenon.

Our third main result concerns weighted sum taken over overpartitions of n and weights involving the number of non-overlined parts of the overpartition.

Theorem 1.7. *For every positive integer n and non-negative integer ℓ, we have*

$$\sum_{\pi\in\overline{P}(n)} (-1)^{\#_{no}(\pi)}\overline{\text{mex}}^{\ell}(\pi) = \begin{cases} (-1)^k((1+k)^\ell - k^\ell) & \textit{if } n = \frac{k(k+1)}{2} \geq 1, \\ 0 & \textit{otherwise}. \end{cases}$$

The $\ell = 1$ and $\ell = 2$ cases of the above theorem are presented as corollaries below.

Corollary 1.8. *For every positive integer n, we have*

$$\sum_{\pi\in\overline{P}(n)} (-1)^{\#_{no}(\pi)}\overline{\text{mex}}(\pi) = \begin{cases} (-1)^k & \textit{if } n = \frac{k(k+1)}{2} \geq 1, \\ 0 & \textit{otherwise}. \end{cases}$$

Corollary 1.9. *For every positive integer n, we have*

$$\sum_{\pi\in\overline{P}(n)} (-1)^{\#_{no}(\pi)}\overline{\text{mex}}^{2}(\pi) = \begin{cases} (-1)^k(2k+1) & \textit{if } n = \frac{k(k+1)}{2} \geq 1, \\ 0 & \textit{otherwise}. \end{cases}$$

Our final main result concerns weighted sum taken over overpartitions of n and weights involving the number of overlined parts of the overpartition.

Theorem 1.10. *For every positive integer n and non-negative integer ℓ, we have*

$$\sum_{\pi\in\overline{P}(n)} (-1)^{\#_{o}(\pi)}\overline{\text{mex}}^{\ell}(\pi) = \begin{cases} (1+k)^\ell - k^\ell & \textit{if } n = \frac{k(k+1)}{2} \geq 1, \\ 0 & \textit{otherwise}. \end{cases}$$

The $\ell = 1$ and $\ell = 2$ cases of the above theorem are presented as corollaries below.

Corollary 1.11. *For every positive integer n, we have*

$$\sum_{\pi\in\overline{P}(n)} (-1)^{\#_{o}(\pi)}\overline{\text{mex}}(\pi) = \begin{cases} 1 & \textit{if } n = \frac{k(k+1)}{2} \geq 1, \\ 0 & \textit{otherwise}. \end{cases}$$

Corollary 1.12. *For every positive integer n, we have*

$$\sum_{\pi\in\overline{P}(n)}(-1)^{\#_o(\pi)}\overline{\text{mex}}^2(\pi)=\begin{cases}2k+1 & \textit{if } n=\frac{k(k+1)}{2}\geq 1,\\ 0 & \textit{otherwise}.\end{cases}$$

To appreciate the significance of our general theorems, similar observations as made for Corollary 1.6, can also be drawn for Corollaries 1.8 and 1.9 and Corollaries 1.11 and 1.12.

2. Proof of Main Theorems

We begin by providing the generating functions for each of the arithmetic functions in Equations (1.7)–(1.10). Using standard combinatorial arguments we can deduce that

$$\sum_{n=1}^{\infty}\sigma_d\text{mex}(c,z,n)q^n=(-cq;q)_\infty\sum_{n=1}^{\infty}\frac{c^{n-1}z^nq^{\frac{n(n-1)}{2}}}{(-cq;q)_n},\tag{2.1}$$

$$\sum_{n=1}^{\infty}\sigma\overline{\text{mex}}(c,z,n)q^n=\frac{(-cq;q)_\infty}{(cq;q)_\infty}\sum_{n=1}^{\infty}c^{n-1}z^n(1-cq^n)q^{\frac{n(n-1)}{2}},\tag{2.2}$$

$$\sum_{n=1}^{\infty}\sigma_{no}\overline{\text{mex}}(c,z,n)q^n=\frac{(-q;q)_\infty}{(cq;q)_\infty}\sum_{n=1}^{\infty}c^{n-1}z^n(1-cq^n)q^{\frac{n(n-1)}{2}},\tag{2.3}$$

and

$$\sum_{n=1}^{\infty}\sigma_o\overline{\text{mex}}(c,z,n)q^n=\frac{(-cq;q)_\infty}{(q;q)_\infty}\sum_{n=1}^{\infty}z^n(1-q^n)q^{\frac{n(n-1)}{2}}.\tag{2.4}$$

Proof of Theorem 1.3. We first equate the right-hand side of (12.2) and (16.3) of Fine [11] to arrive at

$$\frac{1}{(1-t)}\sum_{n=0}^{\infty}\frac{(b/a;q)_n}{(bq;q)_n(tq;q)_n}(-at)^nq^{\frac{n(n+1)}{2}}=\frac{(aq;q)_\infty}{(bq;q)_\infty}\sum_{n=0}^{\infty}\frac{(b/a;q)_n}{(q;q)_n(1-tq^n)}(aq)^n.$$

Then letting $b=0$, $t=-c$, and $a=zq^{-1}$, we get

$$\sum_{n=0}^{\infty}\frac{c^{n-1}z^nq^{\frac{n(n-1)}{2}}}{(-cq;q)_n}=\frac{(z;q)_\infty(1+c)}{c}\sum_{n=0}^{\infty}\frac{z^n}{(q;q)_n(1+cq^n)}.$$

Using this in (2.1), we obtain

$$\begin{aligned}\sum_{n=1}^{\infty}\left(\sum_{\pi\in D(n)}c^{\#(\pi)}z^{\text{mex}(\pi)}\right)q^n&=(-cq;q)_\infty\sum_{n=1}^{\infty}\frac{c^{n-1}z^nq^{\frac{n(n-1)}{2}}}{(-cq;q)_n}\\&=(-cq;q)_\infty\left(\sum_{n=0}^{\infty}\frac{c^{n-1}z^nq^{\frac{n(n-1)}{2}}}{(-cq;q)_n}-\frac{1}{c}\right)\\&=(-cq;q)_\infty\left(\frac{(z;q)_\infty(1+c)}{c}\sum_{n=0}^{\infty}\frac{z^n}{(q;q)_n(1+cq^n)}-\frac{1}{c}\right)\\&=\frac{(-cq;q)_\infty}{c}\left((z;q)_\infty-1+(z;q)_\infty(1+c)\sum_{n=1}^{\infty}\frac{z^n}{(q;q)_n(1+cq^n)}\right)\end{aligned}$$

$$= \frac{(-cq;q)_\infty}{c}\left((z;q)_\infty - 1 + z\sum_{n=1}^{\infty}(1-(-1)^n c^n)(z;q)_{n-1}q^{n-1}\right),$$

where the last line follows from a result of the second author [14, Equation 1.18].

In the case when $c=-1$, we get

$$\sum_{n=1}^{\infty}\left(\sum_{\pi\in D(n)}(-1)^{\#(\pi)}z^{\mathrm{mex}(\pi)}\right)q^n = (q;q)_\infty\left(1-(z;q)_\infty\right).$$

Finally, differentiating both sides with respect to z at $z=1$, we get

$$\sum_{n=1}^{\infty}\left(\sum_{\pi\in D(n)}(-1)^{\#(\pi)}\mathrm{mex}(\pi)\right)q^n = (q;q)_\infty^2.$$

□

Proof of Theorem 1.5. Differentiating both sides of (2.2) with respect to z, we get

$$\sum_{n=1}^{\infty}\left(\sum_{\pi\in\overline{P}(n)}c^{\#(\pi)}\overline{\mathrm{mex}}(\pi)z^{\overline{\mathrm{mex}}(\pi)-1}\right)q^n = \frac{(-cq;q)_\infty}{(cq;q)_\infty}\sum_{n=1}^{\infty}c^{n-1}nz^{n-1}(1-cq^n)q^{\frac{n(n-1)}{2}}.$$

Multiplying both sides by z, then differentiating again with respect to z, and then iterating the same process $\ell-1$ times, we obtain

$$\sum_{n=1}^{\infty}\left(\sum_{\pi\in\overline{P}(n)}c^{\#(\pi)}\overline{\mathrm{mex}}^{\ell}(\pi)z^{\overline{\mathrm{mex}}(\pi)-1}\right)q^n = \frac{(-cq;q)_\infty}{(cq;q)_\infty}\sum_{n=1}^{\infty}c^{n-1}n^{\ell}z^{n-1}(1-cq^n)q^{\frac{n(n-1)}{2}}.$$

Then setting $c=-1$ and $z=1$, we have

$$\begin{aligned}
&\sum_{n=1}^{\infty}\left(\sum_{\pi\in\overline{P}(n)}(-1)^{\#(\pi)}\overline{\mathrm{mex}}^{\ell}(\pi)\right)q^n\\
&= \frac{(q;q)_\infty}{(-q;q)_\infty}\sum_{n=1}^{\infty}(-1)^{n-1}n^{\ell}(1+q^n)q^{\frac{n(n-1)}{2}}\\
&= \frac{(q;q)_\infty}{(-q;q)_\infty}\left(\sum_{n=1}^{\infty}(-1)^{n-1}n^{\ell}q^{\frac{n(n-1)}{2}} + \sum_{n=1}^{\infty}(-1)^{n-1}n^{\ell}q^{\frac{n(n+1)}{2}}\right)\\
&= \frac{(q;q)_\infty}{(-q;q)_\infty}\left(1+\sum_{n=1}^{\infty}(-1)^{n}((n+1)^{\ell}-n^{\ell})q^{\frac{n(n+1)}{2}}\right)\\
&= \left(1+2\sum_{m=1}^{\infty}(-1)^m q^{m^2}\right)\left(1+\sum_{n=1}^{\infty}(-1)^{n}((n+1)^{\ell}-n^{\ell})q^{\frac{n(n+1)}{2}}\right),
\end{aligned}$$

where in the last step we have used (1.2). Finally, comparing the coefficients of q^n on both sides above, we have a proof of the theorem. □

Proof of Theorem 1.7. Applying the operator

$$D_\ell := \left(z\frac{\partial}{\partial z}\right)^\ell \tag{2.5}$$

on (2.3), we get

$$\sum_{n=1}^{\infty}\left(\sum_{\pi\in\overline{P}(n)} c^{\#_{no}(\pi)}\overline{\text{mex}}^\ell(\pi) z^{\overline{\text{mex}}(\pi)-1}\right)q^n = \frac{(-q;q)_\infty}{(cq;q)_\infty}\sum_{n=1}^{\infty} c^{n-1}n^\ell z^{n-1}(1-cq^n)q^{\frac{n(n-1)}{2}}.$$

Then setting $c=-1$ and $z=1$, we get

$$\begin{aligned}\sum_{n=1}^{\infty}\left(\sum_{\pi\in\overline{P}(n)} (-1)^{\#_{no}(\pi)}\overline{\text{mex}}^\ell(\pi)\right)q^n &= \sum_{n=1}^{\infty}(-1)^{n-1}n^\ell(1+q^n)q^{\frac{n(n-1)}{2}} \\ &= \sum_{n=1}^{\infty}(-1)^{n-1}n^\ell q^{\frac{n(n-1)}{2}} + \sum_{n=1}^{\infty}(-1)^{n-1}n^\ell q^{\frac{n(n+1)}{2}} \\ &= 1 - \sum_{n=1}^{\infty}(-1)^{n-1}(n+1)^\ell q^{\frac{n(n+1)}{2}} + \sum_{n=1}^{\infty}(-1)^{n-1}n^\ell q^{\frac{n(n+1)}{2}} \\ &= 1 + \sum_{n=1}^{\infty}(-1)^n((n+1)^\ell - n^\ell)q^{\frac{n(n+1)}{2}}.\end{aligned}$$

Comparing the coefficients of q^n on both sides above, we have our theorem. □

Proof of Theorem 1.10. Again, similar to the proof above, applying the operator (2.5) on (2.4), we get

$$\sum_{n=1}^{\infty}\left(\sum_{\pi\in\overline{P}(n)} c^{\#_{o}(\pi)}\overline{\text{mex}}^\ell(\pi) z^{\overline{\text{mex}}(\pi)-1}\right)q^n = \frac{(-cq;q)_\infty}{(q;q)_\infty}\sum_{n=1}^{\infty} n^\ell z^{n-1}(1-q^n)q^{\frac{n(n-1)}{2}}$$

Then setting $c=-1$ and $z=1$, we get

$$\begin{aligned}\sum_{n=1}^{\infty}\left(\sum_{\pi\in\overline{P}(n)} (-1)^{\#_{o}(\pi)}\overline{\text{mex}}^\ell(\pi)\right)q^n &= \sum_{n=1}^{\infty}n^\ell(1-q^n)q^{\frac{n(n-1)}{2}} \\ &= \sum_{n=1}^{\infty}n^\ell q^{\frac{n(n-1)}{2}} - \sum_{n=1}^{\infty}n^\ell q^{\frac{n(n+1)}{2}} \\ &= \sum_{n=0}^{\infty}(n+1)^\ell q^{\frac{n(n+1)}{2}} - \sum_{n=1}^{\infty}n^\ell q^{\frac{n(n+1)}{2}} \\ &= \sum_{n=0}^{\infty}((n+1)^\ell - n^\ell)q^{\frac{n(n+1)}{2}}.\end{aligned}$$

Comparing the coefficients of q^n on both sides above, we have a proof of the theorem. □

3. Discussions and future work

Similar refinements as discussed in this paper can be pursued for other minimal excludant functions which exist in literature with the techniques used here. For instance, one could consider the overpartition minimal excludant functions defined by the first, third and fourth authors in [10], whose generating functions defined on four different classes of restricted overpartitions give rise to two fifth order mock theta functions, the function $R(q)$ defined earlier and its companion function

$$F(q) = \sum_{n=1}^{\infty} \frac{(-1)^n q^{n^2}}{(q;q^2)_n}.$$

The last two results in the aforementioned paper concern with two sum of mex functions where the minimal excludant is defined on two restricted classes of overpartitions from the four mentioned above. One could also consider other restricted partition classes. However, not all considerations lead to new or interesting results such as the closed forms obtained in our theorems. For instance, the keen observer might notice that we do not consider a similar refinement of the weighted sum of minimal excludant function due to Andrews and Newman in (1.3) because if we did, then with certain substitutions, such as with $c = 1$, we would end up reproducing Jacobi's identity which we describe here in brief. The generating function for (1.6) is given by

$$\sum_{n=1}^{\infty} \sigma\text{mex}(c,z,n) q^n = \frac{1}{(cq;q)_\infty} \sum_{n=1}^{\infty} c^{n-1} z^n (1-cq^n) q^{\frac{n(n-1)}{2}}. \tag{3.1}$$

Differentiating (3.1) with respect to z and setting $z = -1$, $c = 1$, we obtain

$$\sum_{n=1}^{\infty} \left(\sum_{\pi \in P(n)} (-1)^{\text{mex}(\pi)-1} \text{mex}(\pi) \right) q^n = \frac{1}{(q;q)_\infty} \sum_{n=1}^{\infty} (-1)^{n-1} n (1-q^n) q^{\frac{n^2-n}{2}}. \tag{3.2}$$

The sum on the left-hand side of (3.2) is $(q;q)_\infty^2$ as seen from (1.11). This gives

$$(q;q)_\infty^3 = \sum_{n=1}^{\infty} (-1)^{n-1} n (1-q^n) q^{\frac{n^2-n}{2}} = \sum_{n=0}^{\infty} (-1)^n (2n+1) q^{\frac{n^2+n}{2}}, \tag{3.3}$$

where the second equality of (3.3) follows via a change of summation index. Another instance is setting $c = -1$ and $z = 1$ in the second equality of (2.3), we get

$$\sum_{n=1}^{\infty} \left(\sum_{\pi \in \overline{P}(n)} (-1)^{\#_{no}(\pi)} \right) q^n = \sum_{n=1}^{\infty} (-1)^{n-1} (1+q^n) q^{n(n-1)/2} = 1. \tag{3.4}$$

Comparing the coefficients of q^n on both sides of (3.4), we conclude that *for every positive integer n, the number of overpartitions of n having an even number of non-overlined parts is equal to the number of overpartitions of n having an odd number of non-overlined parts*. Similarly, setting $c = -1$ and $z = 1$ in the second equality of (2.4), we get

$$\sum_{n=1}^{\infty} \left(\sum_{\pi \in \overline{P}(n)} (-1)^{\#_o(\pi)} \right) q^n = \sum_{n=1}^{\infty} (1-q^n) q^{n(n-1)/2} = 1. \tag{3.5}$$

Comparing the coefficients of q^n on both sides of (3.5), we arrive at the analogous simple result that *for every positive integer n, the number of overpartitions of n having an even number of overlined* *parts is equal to the number of overpartitions of n having an odd number of overlined parts*. Finally, setting $c = -1$ and $z = 1$ in the second equality of (2.2), we get

$$\sum_{n=1}^{\infty}\left(\sum_{\pi\in\overline{P}(n)}(-1)^{\#(\pi)}\right)q^n = \frac{(q;q)_\infty}{(-q;q)_\infty}\sum_{n=1}^{\infty}(-1)^{n-1}(1+q^n)q^{n(n-1)/2}$$

$$= \frac{(q;q)_\infty}{(-q;q)_\infty} = 1 + 2\sum_{n=1}^{\infty}(-1)^n q^{n^2}, \tag{3.6}$$

where (3.6) follows from (1.2). Comparing the coefficients of q^n on both sides of (3.6) gives us

$$\overline{p}_e(n) - \overline{p}_o(n) = \begin{cases} 2(-1)^k & \text{if } n = k^2, \\ 0 & \text{otherwise,} \end{cases} \tag{3.7}$$

where $\overline{p}_e(n)$ and $\overline{p}_o(n)$ denote the number of overpartitions of n having an even and odd number of parts respectively. Equation (3.7) obtained analytically above is same as Theorem 1.2, described by Andrews in the language of restricted partitions before overpartitions were formally discovered and defined.

Finally, combinatorial proofs to our results, Franklin type or otherwise, are highly desirable. We end with an example illustrating Corollary 1.4 in the table below. Adding the values in the last two columns multiplied with their respective mexes in the preceding fourth column gives

$$\sum_{\pi\in D(6)}(-1)^{\#(\pi)}\mathrm{mex}(\pi) = \sum_{\pi\in P(6)}(-1)^{\mathrm{mex}(\pi)-1}\mathrm{mex}(\pi) = -2.$$

Partition π of 6	$\pi \in \mathcal{D}(6)$	$\#(\pi)$	mex(π)	$(-1)^{\#(\pi)}$	$(-1)^{\mathrm{mex}(\pi)-1}$
6	✓	1	1	-1	1
$5+1$	✓	2	2	1	-1
$4+2$	✓	2	1	1	1
$4+1+1$			2		-1
$3+3$			1		1
$3+2+1$	✓	3	4	-1	-1
$3+1+1+1$			2		-1
$2+2+2$			1		1
$2+2+1+1$			3		-1
$2+1+1+1+1$			3		-1
$1+1+1+1+1+1$			2		-1

TABLE 1. Illustrating Corollary 1.4 for partitions of 6.

REFERENCES

[1] K. Alladi, A combinatorial study and comparison of partial theta identities of Andrews and Ramanujan, *Ramanujan J.* **23** (2010) 227–241.

[2] G. E. Andrews, Two theorems of Gauss and allied identities proved arithmetically, Pacific J. Math. **41** (1972), 563–578.

[3] G. E. Andrews, Ramanujan's Lost Notebook V. Euler's partition identity, *Adv. in Math.* **61** (1986), no. 2, 156–164.

[4] G. E. Andrews, F. J. Dyson and D. R. Hickerson, Partitions and indefinite quadratic forms, *Invent. Math.* **91** (1988), no. 3, 391–407.
[5] G. E. Andrews and D. Newman, Partitions and the minimal excludant, *Ann. Comb.* **23** (2019), 249–254.
[6] V. M. R. Aricheta and J. A. L. Donato, A minimal excludant over overpartitions, *J. Integer Seq.* **27** (2024), no. 7, Art. 24.7.1, 20 pp.
[7] N. D. Baruah et al., A refinement of a result of Andrews and Newman on the sum of minimal excludants, *Ramanujan J.* **62** (2023), no. 4, 1045–1067.
[8] S. Corteel and J. Lovejoy, Overpartitions, *Trans. Amer. Math. Soc.* **356**, 2004.
[9] A. Dhar, A. Mukhopadhyay and R. Sarma, Generalization of the extended minimal excludant of Andrews and Newman, *J. Integer Seq.* **26** (2023), Article no. 23.3.7, 19 pages.
[10] A. Dhar, A. Mukhopadhyay and R. Sarma, On new minimal excludants of overpartitions related to some q-series of Ramanujan, *Bull. Austral. Math. Soc.* **112(1)** (2025), 1–13.
[11] N. J. Fine, Basic hypergeometric series and applications, Mathematical Surveys and Monographs, 27, Amer. Math. Soc., Providence, RI, 1988.
[12] C. F. Gauss, Werke, Bd. 3, Koniglichen Gesellschaft der Wissenschaften, Gottingen, 1866.
[13] P. J. Grabner, A. Knopfmacher, Analysis of some new partition statistics, *Ramanujan J.* **12** (2006) 439–454.
[14] R. Gupta, On sum-of-tails identities, *J. Combin. Theory Ser. A* **184** (2021).
[15] B. Hopkins, J. A. Sellers, A. J. Yee, Combinatorial perspectives on the crank and mex partition statistics, *Electron. J. Combin.* **29**, Article P2. 11, 2022.
[16] P. S. Kaur et al., Minimal excludant over partitions into distinct parts, *Int. J. Number Theory* **18** (2022), no. 9, 2015–2028.
[17] S. Ramanujan, The lost notebook and other unpublished papers, Narosa, New Delhi, 1988.

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF FLORIDA, GAINESVILLE, FL 32611, USA
Email address: aritramdhar@ufl.edu

DEPARTMENT OF MATHEMATICS, INDIAN INSTITUTE OF TECHNOLOGY JODHPUR, JODHPUR, RAJASTHAN, 342030, INDIA
Email address: rgupta@iitj.ac.in

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF FLORIDA, GAINESVILLE, FL 32611, USA
Email address: mukhopadhyay.avi@ufl.edu

DEPARTMENT OF MATHEMATICS, THE PENNSYLVANIA STATE UNIVERSITY, UNIVERSITY PARK, PA 16802, USA
Email address: rishabh.sarma@psu.edu